

CO4 - ASSESSMENT TOOL - I

CLASS TEST

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BE CSE

1) ARCHITECTURAL PATTERNS AND MVC

An architectural pattern is a reusable structure for organizing components of an object-oriented system.

Two common architectural patterns:

Model-View-Controller (MVC)

Layered Architecture

MVC consists of:

Model - manages data and business information

View - displays information to the user

Controller - handles user requests

Benefits:

MVC provides separation of concern, improves modularity, maintainability, reusability, and testing.

2) DAO PATTERN

DAO (Data Access Object) is a design pattern that separates database operations from application logic.

TWO ADVANTAGES :

1) Separation of concerns 2) Easier maintenance

WORKING :

Application Logic → DAO → Database

The DAO handles operations such as insert, update, delete, and retrieve, which application logic focuses on the required functionality.

3) OBJECT - ORIENTED DESIGN GUIDELINES & LSP

Object-oriented design guidelines are principles used to create software that is reusable, maintainable, and flexible.

TWO CATEGORIES :

1) Class / Object design guidelines

2) Relationship / Inheritance design guidelines

Liskov Substitution Principle (LSP):

Objects of a subclass should be usable

wherever objects of the superclass are expected without affecting program correctness.

Importance: LSP improves reusability, polymorphism, maintainability, and reliability.

4) DATABASE PERSISTENCE AND ORM

Database persistence means storing data permanently so that it remains available after the application ends.

Two persistence mechanisms:

1) ORM 2) Direct database access using SQL

ORM (Object-Relational Mapping) maps:

- Class $\rightarrow$ Table
- Object $\rightarrow$ Row
- Attribute $\rightarrow$ Column

ORM helps store, retrieve, update and delete objects in a database with less manual SQL.

5) APPLICATION LAYER

The application layer coordinates application-specific tasks and use cases.

TWO RESPONSIBILITIES:

- 1) Coordinate use cases
- 2) Manage application workflow

WORKING:

Presentation Layer → Application Layer →
Domain Layer

The Application layer coordinates the required operations do not implement domain business rules.

Business rules are handled by Domain/
Business layer.

This separation improves maintainability
and clear responsibility.